Package 'rENM.reports'

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Type Package

Title Report generation tools for the rENM framework

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Description The rENM framework is a modular sysem for
reconstucting and analyzing long-term ecological niche
dynamics using time-indexed environmental data and species
occurrence records. Its packages support climate-based
niche modeling, temporal analysis of climatic suitability,
and reproducible workflows for studying ecological change.

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Encoding UTF-8

RoxygenNote 7.3.3

URL https://github.com/rENM-Framework/rENM.reports

BugReports https://github.com/rENM-Framework/rENM.reports/issues

Imports rENM.core,
dplyr,
gt,
magick,
officer,
openxlsx,
pdftools,
png,
readr

Suggests chromote,
pagedown,
webshot2

Contents

`assemble_centroid_trends_page`

Assemble a "Centroid Trends" page

Description

Builds a single-page PDF combining a suitability map with centroids and a centroid trend summary table.

Usage

```
assemble_centroid_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  margin_top_in = 1,
  margin_right_in = 1,
  margin_bottom_in = 1,
  margin_left_in = 1,
  dpi = 300,
  table_dpi = 600,
  map_width_fraction = 1,
  table_width_fraction = 1,
  v_gap_in = 0.25
)
```

Arguments

`alpha_code` Character. Species alpha code (case-insensitive).

`page_width_in` Numeric. Page width in inches. Default 8.5.

`page_height_in` Numeric. Page height in inches. Default 11.

`margin_top_in` Numeric. Top margin in inches. Default 1.0.

`margin_right_in`
Numeric. Right margin in inches. Default 1.0.

`margin_bottom_in`
Numeric. Bottom margin in inches. Default 1.0.

margin_left_in Numeric. Left margin in inches. Default 1.0.

dpi Integer. Working DPI for the page canvas and PNG scaling. Default 300.

table_dpi Integer. Rasterization DPI for the table PDF. Default 600.

map_width_fraction
Numeric. Fraction (0, 1] of content width used for the top map (1.0 = full content width). Default 1.0.

table_width_fraction
Numeric. Fraction (0, 1] of content width used for the table (1.0 = full content width). Default 1.0.

v_gap_in Numeric. Vertical gap in inches between the map and table. Default 0.25.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function assembles precomputed visualization outputs into a single standardized reporting page. It uses only package code and standard R functions and does not source external scripts.

Inputs Required input files:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-With-Centroi
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Centroid-Trend-Summary.pdf
```

Outputs Output file:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Centroid-Trends.pdf
```

Logging Appends a standard eBird-format processing summary to <project_dir>/runs/<alpha_code>/_log.txt (blank line before dashed separator, no trailing separator).

Layout and scaling

- Top: (A) <ALPHA>-Suitability-Trend-With-Centroids.png
- Bottom: (B) <ALPHA>-Centroid-Trend-Summary.pdf
- Map scaled to map_width_fraction x content width
- Table scaled to table_width_fraction x content width
- Both preserve aspect ratio and are centered horizontally.

Methods Image composition is performed using the magick package. The summary table PDF is rasterized using pdftools::pdf_convert() at the specified DPI before scaling and placement.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if any required input is missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Appends a processing summary to the run log file.

Examples

```
## Not run:
assemble_centroid_trends_page("CASP")

## End(Not run)
```

`assemble_final_report` *Assemble a final report for an rENM run*

Description

Combines a predefined or user-specified set of per-page PDF products into a single report PDF for a given species run. Optionally rasterizes the pages and draws centered page numbers on all but the cover page.

Usage

```
assemble_final_report(
  alpha_code,
  pages = NULL,
  page_numbers = TRUE,
  dpi = 150,
  cleanup = TRUE,
  verbose = TRUE
)
```

Arguments

`alpha_code`	Character. Four-letter species alpha code.
`pages`	Character vector. Optional. Specifies which pages to include and their order. May be either: • full filenames (e.g., "BETH-Suitability-Trends.pdf"), or • short keys (see Details; e.g., "Suitability-Trends") If `NULL` (default), the standard page set and order are used.
`page_numbers`	Logical. Add page numbers (default TRUE).
`dpi`	Numeric. Rasterization resolution when numbering.
`cleanup`	Logical. Remove temporary files.
`verbose`	Logical. Print progress messages.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

By default, a fixed set of pages is assembled in a predefined order. Users may override this behavior via the `pages` argument to:

- reorder pages
- omit pages

- include a subset of pages

Inputs

- Pre-generated page PDFs in <rENM_project_dir()>/runs/<alpha_code>/Summaries/pages/

Outputs

- Final report: <rENM_project_dir()>/runs/<alpha_code>/Summaries/ <alpha_code>-Final-Report.pdf
- Processing summary appended to <rENM_project_dir()>/runs/<alpha_code>/_log.txt

Value

Character. Invisibly returns the final PDF path.

Examples

```
## Not run:
assemble_final_report("BETH")

# Reorder and omit pages
assemble_final_report(
  "BETH",
  pages = c(
    "Suitability-Trend-Analysis",
    "Suitability-Trends",
    "Centroid-Trends"
  )
)

## End(Not run)
```

`assemble_range_timeseries_page` *Assemble a "Range Time Series" page*

Description

Builds a single-page PDF that vertically stacks a collection of range maps above a shared caption with controlled spacing and proportional scaling.

Usage

```
assemble_range_timeseries_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_in = 1,
  target_content_width_in = 6,
  map_width_fraction = 0.85,
  caption_width_fraction = 0.85,
  gap_between_in = 0.5,
```

```
  trim_fuzz = 5,
  verbose = TRUE
)
```

Arguments

- **alpha_code**: Character. Species alpha code (e.g., "CASP"). Converted to uppercase internally for file naming and paths.
- **page_width_in**: Numeric. Page width in inches; default 8.5 (Letter).
- **page_height_in**: Numeric. Page height in inches; default 11 (Letter).
- **dpi**: Numeric. Raster resolution (dots per inch) for caption rendering and final layout. Default 300.
- **margin_in**: Numeric. Uniform page margin in inches. Default 1.
- **target_content_width_in**: Numeric. Target graphic width in inches when both width fractions equal 1.0. Default 6.
- **map_width_fraction**: Numeric. Fraction (0,1] of target_content_width_in assigned to the map. Default 0.85.
- **caption_width_fraction**: Numeric. Fraction (0,1] of target_content_width_in assigned to the caption. Default 0.85.
- **gap_between_in**: Numeric. Vertical gap between map and caption in inches. Default 0.50; set to 0 for no gap.
- **trim_fuzz**: Numeric. Fuzz percentage for magick::image_trim().
  - 0 trims only perfectly uniform borders.
  - 5-10 often works for anti-aliased white backgrounds.

  Default 5.
- **verbose**: Logical. If TRUE, prints diagnostic messages. Default TRUE.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context

- Input map: per-species PNG created earlier in the reporting workflow.
- Input caption: shared PDF stored in the rENM.reports package.
- Output page: PDF saved under each species' Summaries/pages/ directory.

Inputs

- Range map PNG (per species): <project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Range-M
- Range time-series caption PDF (first page only, shared): system.file("captions", "range_timeseries_capti package = "rENM.reports")

Outputs

- Assembled PDF page: <project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Range-TimeSe
- Log entry appended to: <project_dir>/runs/<alpha_code>/_log.txt

Methods

- Rasterization: `pdftools::pdf_render_page()` at the requested `dpi`.
- Margin trimming: `magick::image_trim()` with `trim_fuzz` tolerance.
- Scaling: width fractions applied to `target_content_width_in * dpi` pixels.
- Gap: white spacer of `gap_between_in * dpi` pixels inserted between map and caption.

Data requirements All file paths must exist and be writable; the species directory created by earlier rENM steps is assumed present.

Value

Invisible `Character`. Full file path of the assembled PDF.

- Structure: length-one character vector.
- Side effects: writes the PDF to disk and appends a processing summary to the species log file.

Examples

```
## Not run:
assemble_range_timeseries_page("CASP")

assemble_range_timeseries_page(
  "CASP",
  map_width_fraction     = 1.0,
  caption_width_fraction = 0.9,
  gap_between_in         = 0.25
)

## End(Not run)
```

`assemble_state_trends_page` *Assemble a "State Trends" page*

Description

Builds a single-page PDF that combines a state-level suitability map and a state hot spot map with a summary table into a structured layout.

Usage

```
assemble_state_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
```

```
  h_gutter_in = 0.2,
  v_gap_in = 1,
  map_width_factor = 1,
  table_width_factor = 0.75,
  table_max_height_in = 2.5,
  prefer_table_pdf = TRUE,
  table_pdf_dpi = 600,
  table_png_supersample = 1.5,
  table_sharpen = TRUE,
  trim_maps = TRUE,
  trim_table = TRUE,
  trim_fuzz = 8,
  append_version_to_filename = FALSE,
  debug_frames = FALSE
)
```

Arguments

`alpha_code` Character. Species code (case-insensitive).

`page_width_in` Numeric. Page width in inches. Default 8.5.

`page_height_in` Numeric. Page height in inches. Default 11.

`dpi` Numeric. DPI for raster composition. Default 300.

`margin_left_in` Numeric. Left margin in inches. Default 0.75.

`margin_right_in` Numeric. Right margin in inches. Default 0.75.

`margin_bottom_in` Numeric. Bottom margin in inches. Default 0.75.

`margin_top_in` Numeric. Top margin in inches. Default 1.00.

`h_gutter_in` Numeric. Horizontal gap between maps in inches. Default 0.20.

`v_gap_in` Numeric. Vertical gap between maps and table in inches. Default 1.00.

`map_width_factor` Numeric. Fraction (0 to 1) of content width used by the map row. Default 1.00.

`table_width_factor` Numeric. Fraction (0 to 1) of content width used by the table. Default 0.75.

`table_max_height_in` Numeric. Maximum table height in inches. Default 2.50.

`prefer_table_pdf` Logical. If TRUE and a PDF exists, read the table using `magick::image_read_pdf()`. Default TRUE.

`table_pdf_dpi` Integer. DPI used when rasterizing table PDF. Default 600.

`table_png_supersample` Numeric. Supersampling factor for PNG tables; values greater than or equal to 1 increase resolution (for example, 2 = 200 percent). Default 1.5.

`table_sharpen` Logical. If TRUE, apply mild sharpening after scaling. Default TRUE.

`trim_maps` Logical. If TRUE, trim borders from map PNGs. Default TRUE.

`trim_table` Logical. If TRUE, trim borders from the table image. Default TRUE.

`trim_fuzz` Integer. Tolerance (0 to 100 percent) for trimming. Default 8.

`append_version_to_filename`
: Logical. If TRUE, append a timestamp to the output filename to avoid preview caching. Default FALSE.

`debug_frames`
: Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function composes multiple precomputed visualization products into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analys
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analys
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.png
 (Optional) PDF table:
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-State-Trends.pdf
```

Layout and composition

- Two top-aligned maps (State Analysis and Hotspots) are placed under a configurable top margin.
- The map row can be narrowed and centered using `map_width_factor`.
- A summary table is placed exactly `v_gap_in` inches below the taller of the two maps.
- The table can be narrowed and centered using `table_width_factor`.
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Table rendering

- If `prefer_table_pdf = TRUE`, the table is read from PDF using `magick::image_read_pdf()` and rasterized at `table_pdf_dpi` (for example, 600) to improve text clarity.
- If using a PNG, `table_png_supersample` (for example, 2 for 200 percent) increases resolution prior to fitting.
- Optional sharpening can be applied using `table_sharpen = TRUE`.

Trimming

- Uniform borders are trimmed from maps and optionally from the table to remove hidden padding.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods All image reading, trimming, scaling, and composition are handled using the magick package. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if required PNGs or table sources are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_state_trends_page("CASP")

## End(Not run)
```

---

assemble_suitability_timeseries_page

Assemble a "Suitability Time Series" page

Description

Builds a single-page PDF that vertically stacks a collection of suitability maps above a shared caption with controlled spacing and proportional scaling.

Usage

```
assemble_suitability_timeseries_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_in = 1,
  target_content_width_in = 6,
  map_width_fraction = 0.85,
  caption_width_fraction = 0.85,
  gap_between_in = 0.5,
  trim_fuzz = 5,
  verbose = TRUE
)
```

Arguments

`alpha_code`	Character. Species alpha code (for example, "CASP"). Converted internally to uppercase for file naming and paths.
`page_width_in`	Numeric. Page width in inches. Default 8.5.
`page_height_in`	Numeric. Page height in inches. Default 11.
`dpi`	Numeric. Target raster resolution in dots per inch for both the caption rendering and the final layout. Default 300.

`margin_in` Numeric. Uniform margin on all sides in inches. Default 1.

`target_content_width_in`
: Numeric. Target graphic width in inches for the map and caption when width fractions are 1.0. Default 6.

`map_width_fraction`
: Numeric. Fraction (0, 1] of target width for map.

`caption_width_fraction`
: Numeric. Fraction (0, 1] of target width for caption.

`gap_between_in` Numeric. Vertical gap between map and caption in inches.

`trim_fuzz` Numeric. Fuzz percentage for `magick::image_trim()`.

`verbose` Logical. If TRUE, print diagnostic messages.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context For a given species alpha code (`alpha_code`), this function builds a single-page PDF that stacks:

1. The suitability contact-sheet map PNG (A) on the top of the page: `<project_dir>/runs/<alpha_code>/Summaries/`
2. A shared suitability time-series caption PDF (B; first page only) underneath: `system.file("captions", "suitability_timeseries_caption.pdf", package = "rENM.reports")`

Both elements are:

- Converted to raster images
- Trimmed to remove uniform white margins via `magick::image_trim()`
- Scaled so that, when `map_width_fraction = 1` and `caption_width_fraction = 1`, the visible width of both (A) and (B) is `target_content_width_in` inches (default 6), subject to the page interior width
- Separated by an adjustable vertical gap of `gap_between_in` inches
- Stacked vertically (map above gap, gap above caption)
- Placed on a white Letter-sized page (8.5 x 11 inches) with uniform margins

Trimming the source rasters first ensures that any extra baked-in white margins are removed, so the final 6-inch width corresponds to the graphic itself rather than its original page padding.

The function also appends an eBird-standard processing summary to `<project_dir>/runs/<alpha_code>/_log.txt`, including the caption source path.

Inputs

```
<project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Suitability-Maps.png
 system.file("captions", "suitability_timeseries_caption.pdf",
             package = "rENM.reports")
```

Outputs

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-TimeSeries.pdf
```

Methods All image reading, trimming, scaling, and composition are handled using the magick package. The caption PDF is rasterized using `pdftools::pdf_render_page()`.

Data requirements Required input files must exist prior to execution. The function will stop with an error if the map PNG or caption PDF is missing.

Value

Character. Invisibly returns the output PDF file path:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-TimeSeries.pdf
```

Side effects:

- Writes a PDF file to the project directory
- Creates output directories if they do not exist
- Appends a processing summary to the run log file

Examples

```
## Not run:
assemble_suitability_timeseries_page("CASP")

## End(Not run)
```

---

assemble_suitability_trends_page

Assemble a "Suitability Trends" page

Description

Builds a single-page PDF combining a suitability trend map, a suitability change trend map (acceleration/deceleration), and a caption.

Usage

```
assemble_suitability_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  margin_top_in = 1,
  margin_right_in = 1,
  margin_bottom_in = 1,
  margin_left_in = 1,
  dpi = 300,
  caption_dpi = 600,
  panel_gap_in = 0.15,
  v_gap_in = 0.25
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
margin_top_in	Numeric. Top margin in inches. Default 1.

`margin_right_in`
: Numeric. Right margin in inches. Default 1.

`margin_bottom_in`
: Numeric. Bottom margin in inches. Default 1.

`margin_left_in`
: Numeric. Left margin in inches. Default 1.

`dpi`
: Integer. Working resolution for page/maps (dots per inch). Default 300.

`caption_dpi`
: Integer. Rasterization DPI for the caption PDF (higher = crisper). Default 600.

`panel_gap_in`
: Numeric. Horizontal gap (inches) between the two top panels. Default 0.15.

`v_gap_in`
: Numeric. Vertical gap (inches) between the top row and the caption. Default 0.25.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Produces a single-page PDF with:

- Top row:
  - (A) "<ALPHA>-Suitability-Trend.png"
  - (B) "<ALPHA>-Suitability-Change-Trend.png", side-by-side.
- Bottom:
  - (C) A fixed, shared caption PDF included with the rENM.reports package: "inst/captions/suitability_trend_capt (accessed via system.file()).

The caption is rasterized via pdftools at high DPI to preserve original typography (including bold), centered horizontally, and never stretched (aspect ratio preserved).

Inputs

- Caption PDF is read directly from installed package resources (no staging).
- A and B are read as PNG files from: `<project_dir>/runs/<alpha_code>/Trends/suitability/`

Processing steps

- Reads the caption PDF directly from the installed package resources (no staging).
- Reads A + B (PNG) and rasterizes C (PDF) using pdftools::pdf_convert() at a specified caption_dpi (default 600).
- Scales the caption down only if wider than the content area; otherwise draws at native size, centered.
- Composites everything onto a Letter page and writes: `<project_dir>/runs/<alpha_code>/Summaries/pages/` "<alpha_code>-Suitability-Trends.pdf".

Outputs

- Output PDF written to: `<project_dir>/runs/<alpha_code>/Summaries/pages/` "<alpha_code>-Suitability-Trends.pdf"

Methods Why pdftools? Using pdftools::pdf_convert() (Poppler) avoids font substitution issues that can occur when ImageMagick/Ghostscript cannot find the bold font family. Rasterizing at a high DPI preserves the visual weight of bold text exactly as in the source PDF. The image is then centered and never stretched (aspect ratio preserved).

Data requirements Dependencies: Requires the magick and pdftools packages. If png is missing on your system, install it as well; pdf_convert() writes a PNG which is then read via magick.

Logging: Appends an eBird-standard processing summary to <project_dir>/runs/<alpha_code>/_log.txt with a blank line before the dashed separator and no trailing separator, including the caption source path.

Value

Character. Invisibly returns the output PDF path.

- Output file written to: <project_dir>/runs/<alpha_code>/Summaries/pages/ "<alpha_code>-Suitability-Trends.pdf"
- Appends processing summary to: <project_dir>/runs/<alpha_code>/ _log.txt

Examples

```
## Not run:
assemble_suitability_trends_page("CASP")
assemble_suitability_trends_page("CASP", caption_dpi = 900)

## End(Not run)
```

`assemble_variable_trends_page` *Assemble a "Variable Trends" page*

Description

Builds a single-page PDF that vertically stacks a variable-contribution lines plot above a summary table with proportional scaling and centered alignment.

Usage

```
assemble_variable_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  v_gap_in = 0.5,
  plot_width_factor = 0.85,
  table_width_factor = 0.5,
  table_max_height_in = 3,
  prefer_table_pdf = TRUE,
  table_pdf_dpi = 600,
  table_png_supersample = 1.5,
  table_sharpen = TRUE,
```

```
  trim_plot = TRUE,
  trim_table = TRUE,
  trim_fuzz = 8,
  append_version_to_filename = FALSE
)
```

Arguments

`alpha_code`	Character. Species code (case-insensitive).
`page_width_in`	Numeric. Page width in inches. Default 8.5.
`page_height_in`	Numeric. Page height in inches. Default 11.
`dpi`	Numeric. DPI for the output page raster composition. Default 300.
`margin_left_in`	Numeric. Left margin in inches. Default 0.75.
`margin_right_in`	Numeric. Right margin in inches. Default 0.75.
`margin_bottom_in`	Numeric. Bottom margin in inches. Default 0.75.
`margin_top_in`	Numeric. Top margin in inches. Default 1.00.
`v_gap_in`	Numeric. Visible gap in inches between plot and table. Default 0.50.
`plot_width_factor`	Numeric. Fraction (0, 1] of content width used by the plot. Default 0.85.
`table_width_factor`	Numeric. Fraction (0, 1] of content width used by the table. Default 0.50.
`table_max_height_in`	Numeric. Maximum table height in inches. Default 3.00.
`prefer_table_pdf`	Logical. If TRUE and a PDF exists, read the table using `magick::image_read_pdf()`. Default TRUE.
`table_pdf_dpi`	Integer. DPI when rasterizing the table PDF. Default 600.
`table_png_supersample`	Numeric. Greater than or equal to 1; upscale PNG tables by this factor before fitting (for example, 2 = 200 percent). Default 1.5.
`table_sharpen`	Logical. If TRUE, apply mild sharpening after scaling. Default TRUE.
`trim_plot`	Logical. If TRUE, trim borders from the plot image. Default TRUE.
`trim_table`	Logical. If TRUE, trim borders from the table image. Default TRUE.
`trim_fuzz`	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
`append_version_to_filename`	Logical. If TRUE, append a timestamp to the output filename to avoid preview caching. Default FALSE.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function composes precomputed variable-trend visualizations into a single standardized reporting page. All operations are performed using package-based functionality without sourcing external scripts.

Inputs Required input files:

```
<project_dir>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Lines-
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Variable-Trend-Summary.pdf
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Variable-Trend-Summary.png
```

Outputs Output file:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Variable-Trends.pdf
```

Layout and scaling

- Top: variable-contribution lines plot
- Bottom: summary table
- Plot is placed below the top margin
- Table is placed exactly `v_gap_in` inches below the plot
- Both elements are scaled proportionally and centered horizontally
- No cropping is performed; only scaling

Table rendering

- By default, the table is read from PDF and rasterized at high DPI using `magick::image_read_pdf()` for crisp text
- If PDF is unavailable or `prefer_table_pdf = FALSE`, a PNG is used instead
- PNG tables may be supersampled using `table_png_supersample` (for example, 2 = 200 percent)
- Optional sharpening can be applied using `table_sharpen`

Logging The function prints progress messages and appends an eBird-style processing summary to `<project_dir>/runs/<alpha_code>/_log.txt` with start and stop times and elapsed seconds. The log entry includes a blank line before the separator and no trailing separator after the entry.

Methods All image reading, trimming, scaling, and composition is handled using the magick package. Table rasterization from PDF uses `magick::image_read_pdf()`.

Data requirements Required input files must exist prior to execution. The function will stop with an error if required inputs are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Appends a processing summary to the run log file.

Examples

```
## Not run:
assemble_variable_trends_page("CASP")

## End(Not run)
```

`assemble_variable_trend_maps_page`

Assemble "Variable Trend Maps" pages

Description

Builds a multi-page PDF that shows trends and acceleration/deceleration patterns of the top contributing variables.

Usage

```
assemble_variable_trend_maps_page(alpha_code, overwrite = TRUE, verbose = TRUE)
```

Arguments

`alpha_code`	Character. Species alpha code (e.g., "CASP"). Converted internally to uppercase for all paths and filenames.
`overwrite`	Logical. If TRUE (default) overwrites existing destination files; if FALSE the function stops if a destination file already exists.
`verbose`	Logical. If TRUE (default) prints processing steps and paths to the console.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Pipeline context

- Copies a shared caption PDF (A) and a species-specific variable-trend-maps PDF (B) into the `Summaries/pages` directory for the target species.
- Ensures consistent naming so that downstream report builders pick up both pages automatically.

Inputs

- Shared caption PDF (A):
  `system.file("captions", "variable_trend_maps_caption.pdf", package = "rENM.reports")`
- Species maps PDF (B):
  `file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "maps", paste0(<alpha_code>, "-Variable-Trend-Maps.pdf"))`

Outputs

- Caption copy:
  `file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps1.pdf"))`
- Maps copy:
  `file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps2.pdf"))`
- Processing summary appended to:
  `file.path(rENM_project_dir(), "runs", <alpha_code>, "_log.txt")`

Methods

- Pure file staging via `file.copy()` with optional overwrite.
- No PDF merging, rasterization, or resizing.

Value

A named list (invisibly returned):

- `caption_page` – full file path to the staged caption PDF
- `maps_page` – full file path to the staged maps PDF

Side effects:

- Two PDF files copied into `<project>/runs/<alpha_code>/Summaries/pages`
- Processing summary appended to `<project>/runs/<alpha_code>/_log.txt`

Examples

```
## Not run:
assemble_variable_trend_maps_page("CASP")

## End(Not run)
```

`create_centroid_trend_summary_table`
Create a centroid shift summary table

Description

Description

Usage

```
create_centroid_trend_summary_table(
  alpha_code,
  decimals = 2,
  font_name = "Arial",
  title_size = 10,
  target_px = 350
)
```

Arguments

`alpha_code`	Character. Species alpha code (case-insensitive).
`decimals`	Integer. Number of decimals to display and format.
`font_name`	Character. Table font family.
`title_size`	Integer. Title font size in Excel and PNG/PDF.
`target_px`	Integer. Target pixel width used to center PNG tables.

Details

Build a two-part, publication-ready centroid summary for a species alpha_code. The output combines a shift block and regression block into a standardized tabular format for reporting centroid trends.

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Builds a standardized centroid trend summary from precomputed centroid velocity and regression outputs for a given species.

Inputs Reads CSV files from:

- <project_dir>/runs/<ALPHA>/Trends/centroids/

Expected inputs:

Bioclimatic-Velocity <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Bioclimatic-Velocity.csv with columns start_lon, start_lat, end_lon, end_lat, distance_km, bearing_deg, velocity_km_per_yr.

Longitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Longitude-Summa with columns mean, ci_low, ci_high, pd, rope_pct.

Latitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Latitude-Summary with the same columns as Longitude-Summary.

Outputs Writes outputs to:

- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.xlsx (always written)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.png (if optional graphics packages are available)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.pdf (if optional graphics packages are available)

Table structure

- Shift block (one row): 1980/2020 centroid coordinates, great-circle distance (km), bearing (deg), velocity (km/yr).
- Regression block (two rows: Longitude, Latitude): Slope, CI Low, CI High, PD, Rope %.

Visual and typographic specification

- Title centered and bold; default 10 pt (configurable).
- Table text 8 pt (Arial default).
- No top border or rule on any block.
- No vertical lines anywhere.
- No horizontal rules between body rows in the regression block.
- Column-label rows have a light-gray ("#BFBFBF") underline; each block capped with a light-gray bottom rule.
- PNG/PDF export uses **gt** plus **webshot2** or **chromote**, with both top and bottom tables centered using table.width and table.align and canvas padding via **magick**.
- PNG/PDF use 1 px data-row padding.

Logging Prints progress to the console and appends an eBird-style processing summary (72-dash ruler; Timestamp, Alpha code, Outputs saved, Output file, Total elapsed) to: <project_dir>/runs/<ALPHA>/ _log.txt.

Value

Invisibly returns a List with:

- `shift_df`: Data frame containing the one-row shift block.
- `reg_df`: Data frame containing the two-row regression block.
- `files`: Character vector of file paths written.

Side effects:

- Writes Excel, PNG, and PDF files to the project directory.
- Appends a processing summary to the run-specific log file.

Examples

```
## Not run:
create_centroid_trend_summary_table("CASP")
create_centroid_trend_summary_table("GRRO",
  title_size = 11, font_name = "Helvetica")

## End(Not run)
```

`create_suitability_trend_summary_table`
Create a state-level GAP range / hot spot summary table

Description

Produces a publication-ready summary table from state-level suitability trend data, exporting to Excel and optionally to image and PDF formats.

Usage

```
create_suitability_trend_summary_table(alpha_code)
```

Arguments

`alpha_code`	Character. Four-letter species code.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Converts state-level suitability trend summary outputs into formatted tables suitable for reporting, publication, and visualization.

Inputs Input CSV must exist at:

`<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis`

Processing steps

- Reads the summary CSV using `readr::read_csv`.

- Verifies required columns are present.
- Converts numeric columns safely.
- Renames columns for presentation.
- Builds a formatted Excel workbook using `openxlsx`.
- Creates a styled table using `gt`.
- Optionally exports PNG using webshot2 if available.
- Optionally exports PDF using pagedown if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)
- PNG (.png) if webshot2 is available
- PDF (.pdf) if pagedown is available

Table columns are presented as: State, State Area, Range Area, Range %, Positive %, Negative %, Hot Spot Area, Hot Spot %.

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format.

Data requirements Input CSV must contain: state, state_area, range_area, range_pct, pos_pct, neg_pct, hotspot_area, hotspot_pct.

Value

A named list returned invisibly with the following elements:

- xlsx: Character. Absolute path to Excel output file.
- png: Character. Absolute path to PNG file or NA if skipped.
- pdf: Character. Absolute path to PDF file or NA if skipped.

Side effects:

- Writes formatted Excel, PNG, and PDF outputs.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_suitability_trend_summary_table("CASP")

## End(Not run)
```

create_variable_trend_summary_table

Create a variable trend summary table

Description

Produces a publication-ready summary table from variable-trend statistics with significance marks (*, **, ***) and bolding by PD.

Usage

```
create_variable_trend_summary_table(alpha_code)
```

Arguments

alpha_code	Character. Species alpha code.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Converts variable-level Bayesian trend statistics into a compact, formatted summary table suitable for reporting and publication.

Inputs Input file must exist at:

<project_dir>/runs/<alpha_code>/Trends/variables/ <alpha_code>-Variable-Contributions-BR-Stats.cs

Processing steps

- Reads the variable-trend CSV using readr::read_csv.
- Verifies required columns are present.
- Resolves CI column names: slope_ci_lower or slope_ci_low, and slope_ci_upper or slope_ci_high.
- Preserves row order exactly as in the input CSV.
- Assigns significance marks based on pd_slope:
  - *** if pd_slope >= 95
  - ** if 90 <= pd_slope < 95
  - * if 85 <= pd_slope < 90
- Appends significance marks to variable names.
- Converts numeric fields safely.
- Builds a formatted Excel workbook using openxlsx.
- Creates a styled table using gt.
- Optionally exports PNG using webshot2 if available.
- Optionally exports PDF using pagedown if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)

- PNG (.png) if webshot2 is available
- PDF (.pdf) if pagedown is available

Column mapping Output columns correspond to input columns:

- Variable <- Variable
- Points <- n_points
- Slope <- slope_mean
- CI Low <- slope_ci_lower or slope_ci_low
- CI High <- slope_ci_upper or slope_ci_high
- PD <- pd_slope
- ROPE

Formatting rules

- Row order is preserved exactly as in the input CSV.
- Variables with pd_slope >= 85 are bolded.
- Significance marks are appended to variable names.

Log behavior Appends a processing summary to:

`<project_dir>/runs/<alpha_code>/_log.txt`

using the eBird-standard format:

- One blank line before the 72-dash separator.
- No ending separator line.
- Reports all output files written.

Data requirements Input CSV must contain: Variable, n_points, slope_mean, pd_slope, rope_slope_pct, and CI columns (slope_ci_lower or slope_ci_low, slope_ci_upper or slope_ci_high).

Value

A named list returned invisibly with the following elements:

- xlsx: Character. Absolute path to Excel output file.
- png: Character. Absolute path to PNG output or NA if skipped.
- pdf: Character. Absolute path to PDF output or NA if skipped.

Side effects:

- Writes Excel, PNG, and PDF files to disk.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_variable_trend_summary_table("CASP")

## End(Not run)
```

`gather_range_maps` *Assemble 3x3 range map contact sheet*

Description

Builds a 3x3 contact sheet of range maps for a species code by collecting PNG tiles from TimeSeries outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_range_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

`alpha_code`	Character. Species code used in folder and file names.
`margin_in`	Numeric. Page margin on top, left, and right in inches.
`page_width_in`	Numeric. Page width in inches (Letter = 8.5).
`page_height_in`	Numeric. Page height in inches (Letter = 11).
`tile`	Character. Montage tile specification for magick::image_montage.
`trim_fuzz`	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
`dpi`	Numeric. Rendering resolution for raster-based layout.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Assembles retrospective ENM range map outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

`<project_dir>/runs/<alpha_code>/TimeSeries/<year>/model/<alpha_code>-<year>-Range.png`

for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using `magick::image_trim`.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).

- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Range-Maps.png)
- A one-page PDF (<alpha_code>-Range-Maps.pdf)
- A one-page DOCX (<alpha_code>-Range-Maps.docx)

All files are written to:

`<project_dir>/runs/<alpha_code>/Summaries/maps`

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and is centered horizontally.

Data requirements Exactly 9 range PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: `<project_dir>/runs/<alpha_code>/_log.txt`

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_range_maps("BETH")

# For a different alpha_code:
gather_range_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_range_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

`gather_suitability_maps` *Assemble side-by-side suitability maps into paged composites*

Description

Builds a 3x3 contact sheet of suitability maps for a species code by collecting prediction PNGs from TimeSeries outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_suitability_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

`alpha_code`	Character. Species code used in folder and file names.
`margin_in`	Numeric. Page margin on top, left, and right in inches.
`page_width_in`	Numeric. Page width in inches (Letter = 8.5).
`page_height_in`	Numeric. Page height in inches (Letter = 11).
`tile`	Character. Montage tile specification for magick::image_montage.
`trim_fuzz`	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
`dpi`	Numeric. Rendering resolution for raster-based layout.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Assembles retrospective ENM prediction outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

`<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/model/ <alpha_code>-<year>-Prediction.png`

for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using `magick::image_trim`.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).

- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Suitability-Maps.png)
- A one-page PDF (<alpha_code>-Suitability-Maps.pdf)
- A one-page DOCX (<alpha_code>-Suitability-Maps.docx)

All files are written to:

`<project_dir>/runs/<alpha_code>/Summaries/maps`

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and centered horizontally.

Data requirements Exactly 9 prediction PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: `<project_dir>/runs/<alpha_code>/_log.txt`

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_suitability_maps("BETH")

# For a different alpha_code:
gather_suitability_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_suitability_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

gather_suitability_trend_stats
Merge state-level suitability and hot-spot statistics

Description

Reads two per-state CSV files for a species and merges them by two-letter state abbreviation to produce a concise summary table of suitability and hotspot metrics.

Usage

```
gather_suitability_trend_stats(alpha_code)
```

Arguments

alpha_code	Character. Four-letter species code.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().s

Pipeline context Combines state-level suitability trend outputs with hotspot statistics into a unified summary table for downstream reporting and analysis.

Inputs Input files must exist at:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Processing steps

- Reads both input CSV files using readr::read_csv.
- Verifies required columns are present in each dataset.
- Sorts both inputs alphabetically by state abbreviation.
- Merges datasets using a left join on state abbreviation.
- Converts relevant columns to numeric values.
- Assembles a standardized summary table with key metrics.

Outputs Output file is written to:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format:

- Always skips one blank line before the 72-dash separator.
- Does not include an ending separator line.
- Omits all raster-related fields.
- Includes: Timestamp, Alpha code, Outputs saved, Total elapsed, and Output file.

Data requirements

- First file must contain: STATE, GAP.RANGE.AREA, GAP.RANGE.PCT, GAP.RANGE.POS.PCT, and either GAPP.RANGE.NEG.PCT or GAP.RANGE.NEG.PCT.
- Second file must contain: abbr, state_area_km2, hotspot_area_km2, hotspot_pct_of_state.

Value

Invisibly returns a data.frame with the following columns:

- state: Character. Two-letter state abbreviation.
- state_area: Numeric. State area in square kilometers.
- range_area: Numeric. Area of species range.
- range_pct: Numeric. Percent of state occupied by range.
- pos_pct: Numeric. Percent of positive trend area.
- neg_pct: Numeric. Percent of negative trend area.
- hotspot_area: Numeric. Area classified as hotspot.
- hotspot_pct: Numeric. Percent of state as hotspot.

Side effects:

- Writes a summary CSV file to the suitability directory.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
gather_suitability_trend_stats("CASP")

## End(Not run)
```

---

gather_top_variable_trend_maps

Assemble side-by-side variable trend maps into paged composites

---

Description

Pair trend and rate PNGs for each variable, build A | B montages, arrange several montages per page, and export them as PDF, DOCX, and PNG. Designed for variable-contribution reporting in the rENM workflow.

Usage

```
gather_top_variable_trend_maps(
  alpha_code,
  base_dir = c("auto", "m2", "mc"),
  base_home = "~",
  vars_csv_subpath = NULL,
  per_page = 5,
  dpi = 300,
  page_width_in = 8.5,
```

```
  page_height_in = 11,
  margin_in = 1,
  trim_fuzz = 8
)
```

Arguments

`alpha_code`	Character. Species code (e.g., "CASP"). Used for input and output file naming.
`base_dir`	Character. One of "auto", "m2", "mc"; controls which MERRA-2 directories are searched for trend images.
`base_home`	Character. Deprecated. Ignored and retained only for backward compatibility; the project root is taken from `rENM_project_dir()`.
`vars_csv_subpath`	Character or NULL. Optional path (relative to `runs/<alpha_code>`) to the variable list CSV; when NULL the default path described in **Pipeline steps** is used.
`per_page`	Integer. Number of variables (rows) per output page.
`dpi`	Numeric. Rendering resolution in dots per inch.
`page_width_in`	Numeric. Page width in inches.
`page_height_in`	Numeric. Page height in inches.
`margin_in`	Numeric. Page margin in inches on all sides.
`trim_fuzz`	Numeric. Fuzz parameter passed to `magick::image_trim()` (in pixels).

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline steps

- 1. Read the variable list from `<rENM_project_dir()>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Stats.csv` (must contain `Variable`; `pd_slope` is optional).
- 2. For every variable <var> collect two PNGs (case-insensitive, punctuation-agnostic matching):
  - A (trend): `<base>-trends/<var>.png`
  - B (rate) : `<base>-trends-rate/<var>.png`

  If `base_dir` == `"auto"` both `"m2"` and `"mc"` directories are tried.
- 3. Build an A | B montage for each variable with generous spacing.
- 4. Stack `per_page` montages per page.
- 5. Trim, down-scale, and vertically crop overflow.
- 6. Composite onto a white US Letter page (8.5 x 11 in, 1 in margins).
- 7. Export a multipage PDF, a matching DOCX, and page-level PNGs named `-p01`, `-p02`, ...

Highlight rule If the CSV includes `pd_slope`, any variable with `{pd_slope >= 90}` receives a dark-blue strip along the left edge.

Layout and styling

- Each A <-> B pair is slightly down-scaled to add whitespace.
- The gutter between A and B, as well as row spacing, is widened.

- The full column is scaled to 96% of the content width and centred.
- Excess height beyond the content area is cropped from the bottom.

CRAN compliance All hard-coded paths have been removed; the project root is obtained via `rENM_project_dir()`.

Inputs

- Variable CSV (with optional `pd_slope`) as described above.
- Trend and rate PNGs residing in the two `<base>-trends*` folders under `<rENM_project_dir()>/data/merra/`.

Outputs

- `<alpha_code>-Variable-Trend-Maps.pdf`
- `<alpha_code>-Variable-Trend-Maps.docx`
- `<alpha_code>-Variable-Trend-Maps-pXX.png`
- A processing summary appended to `runs/<alpha_code>/_log.txt`

Value

A `list` with the following elements:

- `png`: Character vector of page-level PNG paths.
- `pdf`: Character string giving the PDF path.
- `docx`: Character string giving the DOCX path.

Side effects:

- Files are written to `runs/<alpha_code>/Summaries/maps/`.
- A log block is appended to `runs/<alpha_code>/_log.txt`.

Examples

```
## Not run:
gather_top_variable_trend_maps("CASP")           # auto-detect bases
gather_top_variable_trend_maps("CASP", "m2")     # force "m2"
gather_top_variable_trend_maps("CASP", per_page = 3)

## End(Not run)
```

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